

ECON 320 – Solutions to Practice Problem Set 1

Sets and Functions

Mathematical Foundations for Economic Analysis

Course: ECON 320: Introduction to Mathematical Economics

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1 Full Solutions

1.1 Solutions: Section A

Solution

A1. Set Builder Notation

(a) Even natural numbers less than 20:

$$A = \{x \in \mathbb{N} : x \text{ is even and } x < 20\} = \{2, 4, 6, 8, 10, 12, 14, 16, 18\}. \quad \text{Finite (9 elements).}$$

(b) Real numbers with $x^2 < 16$:

$$B = \{x \in \mathbb{R} : x^2 < 16\} = \{x \in \mathbb{R} : |x| < 4\} = (-4, 4). \quad \text{Infinite.}$$

(c) Prices with $D(P) = 50 - P > 0$ and $P \in [0, 100]$:

$D(P) > 0 \Rightarrow P < 50$. Combined with $P \in [0, 100]$:

$$C = \{P \in [0, 100] : 50 - P > 0\} = [0, 50). \quad \text{Infinite.}$$

Solution

A2. Set Operations

With $U = \{1, \dots, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 5, 6, 7\}$, $C = \{5, 6, 7, 8, 9\}$:

(a) $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$

(b) $A \cap B = \{3, 4, 5\}$

(c) $A \setminus B = \{1, 2\}$ (elements in A not in B)

(d) $B \setminus A = \{6, 7\}$ (elements in B not in A)

(e) $(A \cup B)^c = U \setminus \{1, 2, 3, 4, 5, 6, 7\} = \{8, 9, 10\}$

(f) $A \cap B \cap C = \{3, 4, 5\} \cap \{5, 6, 7, 8, 9\} = \{5\}$

(g) $B \cap C = \{5, 6, 7\}$, so $A \cup (B \cap C) = \{1, 2, 3, 4, 5\} \cup \{5, 6, 7\} = \{1, 2, 3, 4, 5, 6, 7\}$

Check distributivity: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$, $A \cup C = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. $(A \cup B) \cap (A \cup C) = \{1, 2, 3, 4, 5, 6, 7\}$. ✓

Solution

A3. Power Set

$A = \{x, y\}$, so $|A| = 2$ and $|\mathcal{P}(A)| = 2^2 = 4$:

$$\mathcal{P}(A) = \{\emptyset, \{x\}, \{y\}, \{x, y\}\}$$

Now $|\mathcal{P}(A)| = 4$, so $|\mathcal{P}(\mathcal{P}(A))| = 2^4 = 16$.

Solution**A4. Cartesian Product**

$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$ ($|A \times B| = 2 \times 3 = 6$ elements)

$B \times A = \{(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)\}$ ($|B \times A| = 3 \times 2 = 6$ elements)

$A \times B \neq B \times A$: the pairs are ordered differently; e.g. $(1, a) \in A \times B$ but $(1, a) \notin B \times A$ since $1 \notin B$. In general $A \times B = B \times A$ only if $A = B$.

Solution**A5. De Morgan's Law Verification**

$U = \{1, \dots, 8\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$.

Left side: $A \cup B = \{1, 2, 3, 4, 5, 6\}$, so $(A \cup B)^c = \{7, 8\}$.

Right side: $A^c = \{5, 6, 7, 8\}$, $B^c = \{1, 2, 7, 8\}$, so $A^c \cap B^c = \{7, 8\}$.
 $(A \cup B)^c = A^c \cap B^c = \{7, 8\}$. ✓

Solution**A6. Budget Set Intersection**

$A = \{(x_1, x_2) \in \mathbb{R}_+^2 : 2x_1 + 5x_2 \leq 100\}$ is the full budget set (triangle with vertices $(0, 0)$, $(50, 0)$, $(0, 20)$).

$B = \{(x_1, x_2) \in \mathbb{R}_+^2 : x_1 \geq 10\}$ is the closed half-plane to the right of $x_1 = 10$.

$A \cap B = \{(x_1, x_2) \in \mathbb{R}_+^2 : 2x_1 + 5x_2 \leq 100, x_1 \geq 10\}$

Geometrically: the budget triangle *truncated* by the vertical line $x_1 = 10$. It is a quadrilateral with vertices $(10, 0)$, $(50, 0)$, $(0, 20)$... wait — at $x_1 = 10$: $2(10) + 5x_2 \leq 100 \Rightarrow x_2 \leq 16$. So the binding vertices are $(10, 0)$, $(50, 0)$, and $(10, 16)$.

Economic interpretation: affordable bundles that contain *at least 10 units of good 1* (perhaps a minimum consumption requirement or a rationing constraint). This is the feasible set when the consumer is forced to consume at least 10 units of good 1.

Solution**A7. Proof of Distributive Law**

We prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ by element-chasing.

($\subseteq$): Let $x \in A \cap (B \cup C)$. Then $x \in A$ and $x \in B \cup C$.

Case 1: $x \in B$. Then $x \in A$ and $x \in B$, so $x \in A \cap B \subseteq (A \cap B) \cup (A \cap C)$.

Case 2: $x \in C$. Then $x \in A$ and $x \in C$, so $x \in A \cap C \subseteq (A \cap B) \cup (A \cap C)$.

In either case, $x \in (A \cap B) \cup (A \cap C)$.

($\supseteq$): Let $x \in (A \cap B) \cup (A \cap C)$.

Case 1: $x \in A \cap B$. Then $x \in A$ and $x \in B \subseteq B \cup C$, so $x \in A \cap (B \cup C)$.

Case 2: $x \in A \cap C$. Then $x \in A$ and $x \in C \subseteq B \cup C$, so $x \in A \cap (B \cup C)$.

Since $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ and $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$, equality holds. $\square$

Solution**A8. Supremum and Infimum**

$$(a) A = \left\{ \frac{n}{n+1} : n \in \mathbb{N} \right\} = \left\{ \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots \right\}$$

The sequence $n/(n+1)$ is strictly increasing and approaches 1. So $\sup A = 1$ (not attained: $n/(n+1) < 1$ for all n , so $\max A$ does not exist). $\inf A = \min A = \frac{1}{2}$ (attained at $n = 1$).

$$(b) B = (-3, 5]: \inf B = -3 \text{ (not attained, } \min B \text{ DNE); } \sup B = \max B = 5.$$

$$(c) C = \{x \in \mathbb{R} : x^2 < 9\} = (-3, 3): \inf C = -3 \text{ (DNE); } \sup C = 3 \text{ (DNE).}$$

Neither min nor max exists.

$$(d) D = \{x \in \mathbb{R} : x^3 \leq 8\} = (-\infty, 2]: \sup D = \max D = 2. D \text{ is unbounded below: } \inf D = -\infty \text{ (no infimum in } \mathbb{R}).$$

1.2 Solutions: Section B**Solution****B1. Domain and Range**

$$(a) f(x) = \sqrt{9 - x^2}: \text{ need } 9 - x^2 \geq 0, \text{ i.e. } x^2 \leq 9, \text{ i.e. } x \in [-3, 3].$$

Range: $f(x) \geq 0$ and $f(0) = 3$, so range = $[0, 3]$.

$$(b) g(x) = \frac{x-1}{x^2-4} = \frac{x-1}{(x-2)(x+2)}: \text{ need } x \neq \pm 2.$$

Domain = $\mathbb{R} \setminus \{-2, 2\}$. Range: g takes all real values except... at $x = 1$, $g(1) = 0$. Analysis shows range = $\mathbb{R}$ (every value achieved by continuity and the behaviour near the poles).

$$(c) h(x) = \ln(x^2 - 1): \text{ need } x^2 - 1 > 0, \text{ i.e. } |x| > 1.$$

Domain = $(-\infty, -1) \cup (1, \infty)$. As $|x| \rightarrow 1^+$, $\ln(x^2 - 1) \rightarrow -\infty$; as $|x| \rightarrow \infty$, $\ln(x^2 - 1) \rightarrow \infty$. Range = $\mathbb{R} = (-\infty, \infty)$.

$$(d) k(x, y) = \sqrt{x} + \sqrt{y}: \text{ need } x \geq 0 \text{ and } y \geq 0.$$

Domain = $\mathbb{R}_+^2 = [0, \infty)^2$. Range: $k(x, y) \geq 0$ always; for any $z \geq 0$, take $x = z^2$, $y = 0$: $k(z^2, 0) = z$. Range = $[0, \infty)$.

Solution**B2. Injectivity and Surjectivity**

$$(a) f(x) = 3x - 7: \text{ **Bijective.**}$$

Injective: $3x_1 - 7 = 3x_2 - 7 \Rightarrow x_1 = x_2$. ✓

Surjective: for any $y \in \mathbb{R}$, $x = (y + 7)/3$ gives $f(x) = y$. ✓

$$(b) g(x) = x^3 - x = x(x-1)(x+1): \text{ **Neither.**}$$

Not injective: $g(0) = 0 = g(1)$... let's check: $g(0) = 0$, $g(-1) = (-1)(0) = 0$ wait: $g(-1) = (-1)(-2)(0) = 0$. So $g(0) = g(-1) = 0$ with $0 \neq -1$. Not injective.

Not surjective? Actually it IS surjective since $\lim_{x \rightarrow \pm\infty} g(x) = \pm\infty$ and g is continuous, so by IVT every value in $\mathbb{R}$ is achieved. So g is **surjective but not injective**.

$$(c) h: \mathbb{R} \rightarrow \mathbb{R}^+, h(x) = e^x: \text{ **Bijective.**}$$

Injective: $e^{x_1} = e^{x_2} \Rightarrow x_1 = x_2$. ✓

Surjective (onto $\mathbb{R}^+$): for any $y > 0$, $x = \ln y \in \mathbb{R}$ gives $h(x) = e^{\ln y} = y$. ✓

$$(d) k: [0, \pi] \rightarrow [-1, 1], k(x) = \cos x: \text{ **Bijective.**}$$

Injective: $\cos$ is strictly decreasing on $[0, \pi]$, so injective. ✓

Surjective (onto $[-1, 1]$): $k(0) = 1$, $k(\pi) = -1$, continuous, so by IVT every value in $[-1, 1]$ is achieved. ✓

(e) $\ell : \mathbb{R}_+ \rightarrow \mathbb{R}_+$, $\ell(x) = x^3$: **Bijective**.

Injective: x^3 is strictly increasing on $\mathbb{R}_+$. ✓

Surjective: for any $y \geq 0$, $x = y^{1/3} \geq 0$ gives $\ell(x) = y$. ✓

Solution

B3. Inverse Functions

(a) $f(x) = 5x + 3$: Set $y = 5x + 3 \Rightarrow x = \frac{y-3}{5}$.

So $f^{-1}(y) = \frac{y-3}{5}$, domain $\mathbb{R}$.

(b) $g(x) = \frac{x+1}{x-2}$, $x \neq 2$: Set $y = \frac{x+1}{x-2}$.

$y(x-2) = x+1 \Rightarrow xy - 2y = x+1 \Rightarrow x(y-1) = 2y+1 \Rightarrow x = \frac{2y+1}{y-1}$.

So $g^{-1}(y) = \frac{2y+1}{y-1}$, domain $\mathbb{R} \setminus \{1\}$.

(c) $h(x) = e^{2x-1}$: Set $y = e^{2x-1} \Rightarrow \ln y = 2x-1 \Rightarrow x = \frac{\ln y + 1}{2}$.

So $h^{-1}(y) = \frac{\ln y + 1}{2}$, domain $(0, \infty)$.

(d) $k(x) = x^3 + 2$ on $\mathbb{R}$: Set $y = x^3 + 2 \Rightarrow x^3 = y-2 \Rightarrow x = (y-2)^{1/3}$.

So $k^{-1}(y) = (y-2)^{1/3}$, domain $\mathbb{R}$ (cube roots defined for all reals).

Solution

B4. Composition

$f(x) = 2x^2 - 1$, $g(x) = \sqrt{x+3}$ with domain $[-3, \infty)$.

(a) $(g \circ f)(x) = g(f(x)) = \sqrt{(2x^2 - 1) + 3} = \sqrt{2x^2 + 2} = \sqrt{2} \cdot \sqrt{x^2 + 1}$.

Domain: need $f(x) \in [-3, \infty)$, i.e. $2x^2 - 1 \geq -3$, i.e. $x^2 \geq -1$. This holds for all $x \in \mathbb{R}$.

Domain of $g \circ f$ is $\mathbb{R}$.

(b) $(f \circ g)(x) = f(g(x)) = 2(\sqrt{x+3})^2 - 1 = 2(x+3) - 1 = 2x+5$.

Domain: need $x \in [-3, \infty)$ (the domain of g).

Domain of $f \circ g$ is $[-3, \infty)$.

(c) $(g \circ f)(2) = \sqrt{2(4) + 2} = \sqrt{10}$

$(f \circ g)(1) = 2(1) + 5 = 7$

(d) $(g \circ f)(x) = \sqrt{2}\sqrt{x^2 + 1}$ vs. $(f \circ g)(x) = 2x + 5$. These are clearly different functions.

This illustrates that function composition is **not commutative**: $g \circ f \neq f \circ g$ in general.

Solution**B5. Pre-image**

$$f(x) = x^2 - 4.$$

$$(a) f^{-1}(\{0\}) = \{x : x^2 - 4 = 0\} = \{x : x^2 = 4\} = \{-2, 2\}.$$

$$(b) f^{-1}([-4, 0]) = \{x : -4 \leq x^2 - 4 \leq 0\} = \{x : 0 \leq x^2 \leq 4\} = \{x : |x| \leq 2\} = [-2, 2].$$

$$(c) f^{-1}((0, 5]) = \{x : 0 < x^2 - 4 \leq 5\} = \{x : 4 < x^2 \leq 9\} = \{x : 2 < |x| \leq 3\} = [-3, -2) \cup (2, 3].$$

$$(d) f^{-1}(\{-5\}) = \{x : x^2 - 4 = -5\} = \{x : x^2 = -1\} = \emptyset.$$

(No real number has a negative square, so no input maps to -5 .)

Solution**B6. Homogeneity**

$$(a) f(K, L) = 3K^{0.4}L^{0.6}:$$

$$f(tK, tL) = 3(tK)^{0.4}(tL)^{0.6} = 3t^{0.4}K^{0.4}t^{0.6}L^{0.6} = t^1 \cdot 3K^{0.4}L^{0.6} = t^1 f(K, L).$$

Homogeneous of degree 1 (CRS). By Euler's theorem: $K \cdot \frac{\partial f}{\partial K} + L \cdot \frac{\partial f}{\partial L} = 1 \cdot f(K, L).$

$$(b) g(x, y) = x^2 + xy + y^2:$$

$$g(tx, ty) = (tx)^2 + (tx)(ty) + (ty)^2 = t^2x^2 + t^2xy + t^2y^2 = t^2(x^2 + xy + y^2) = t^2g(x, y).$$

Homogeneous of degree 2.

$$(c) h(x, y) = \frac{x^2y}{x^3 + y^3}:$$

$$h(tx, ty) = \frac{(tx)^2(ty)}{(tx)^3 + (ty)^3} = \frac{t^3x^2y}{t^3(x^3 + y^3)} = \frac{x^2y}{x^3 + y^3} = t^0 \cdot h(x, y).$$

Homogeneous of degree 0 (like Walrasian demand: scale-invariant).

$$(d) k(x, y) = x + y + 1:$$

$$k(tx, ty) = tx + ty + 1 = t(x + y) + 1 \neq t^r(x + y + 1) \text{ for any } r \text{ (the constant 1 is unaffected by scaling).}$$

Not homogeneous of any degree.

Solution**B7. Convexity and Concavity**

$$(a) f(x) = x^4: f''(x) = 12x^2 \geq 0 \text{ for all } x, \text{ with equality only at } x = 0.$$

Convex (not strictly convex everywhere, since $f''(0) = 0$, but it is strictly convex in the sense that $f(\lambda x + (1 - \lambda)y) < \lambda f(x) + (1 - \lambda)f(y)$ for $x \neq y$).

$$(b) g(x) = x^3: g''(x) = 6x. \quad g''(x) > 0 \text{ for } x > 0 \text{ (convex) and } g''(x) < 0 \text{ for } x < 0 \text{ (concave). Neither convex nor concave on all of } \mathbb{R}.$$

$$(c) h(x) = -2x^2 + 4x - 1: h''(x) = -4 < 0 \text{ for all } x. \text{ **Strictly concave.}**$$

$$(d) k(x) = |x|: \text{For } x \geq 0, k'' = 0 \text{ (weakly convex); for } x \leq 0, k'' = 0. \text{ Using the definition: } k(\lambda x + (1 - \lambda)y) \leq \lambda k(x) + (1 - \lambda)k(y) \text{ by the triangle inequality.}$$

$$|\lambda x + (1 - \lambda)y| \leq |\lambda x| + |(1 - \lambda)y| = \lambda|x| + (1 - \lambda)|y| \text{ (since } \lambda, 1 - \lambda \geq 0).$$

Convex (weakly, not strictly).

Solution**B8. Monopolist Economic Application****(a)** Profit function:

$$\pi(Q) = \text{Revenue} - \text{Cost} = P(Q) \cdot Q - C(Q) = (100 - 2Q)Q - (Q^2 + 10Q + 50)$$

$$\pi(Q) = 100Q - 2Q^2 - Q^2 - 10Q - 50 = -3Q^2 + 90Q - 50$$

Domain: $Q \geq 0$ and $P(Q) = 100 - 2Q \geq 0 \Rightarrow Q \leq 50$. So domain = $[0, 50]$.**(b)** $\pi(Q) = -3Q^2 + 90Q - 50$ is a downward parabola. Maximum at $Q^* = 15$ (see (d)):

$$\pi_{\max} = -3(225) + 90(15) - 50 = -675 + 1350 - 50 = 625$$

At endpoints: $\pi(0) = -50$, $\pi(50) = -3(2500) + 4500 - 50 = -50$.Range = $[-50, 625]$.**(c)** $\pi''(Q) = -6 < 0$ for all Q . Therefore π is **strictly concave**.Economic meaning: each additional unit of output raises profit by less and less (eventually reducing it). The profit-maximising Q is uniquely determined and is a global maximum.**(d)** First-order condition: $\pi'(Q) = -6Q + 90 = 0 \Rightarrow Q^* = 15$.

$$\pi(15) = -3(225) + 90(15) - 50 = -675 + 1350 - 50 = \mathbf{625}.$$

Maximum profit is **\$625** at output $Q^* = 15$.**(e)** Break-even: $\pi(Q) = 0$:

$$-3Q^2 + 90Q - 50 = 0 \Rightarrow 3Q^2 - 90Q + 50 = 0$$

$$Q = \frac{90 \pm \sqrt{8100 - 600}}{6} = \frac{90 \pm \sqrt{7500}}{6} = \frac{90 \pm 50\sqrt{3}}{6} = 15 \pm \frac{25\sqrt{3}}{3}$$

$$Q_1 = 15 - \frac{25\sqrt{3}}{3} \approx 15 - 14.43 = 0.57, \quad Q_2 = 15 + \frac{25\sqrt{3}}{3} \approx 29.43$$

So $\pi^{-1}(\{0\}) \approx \{0.57, 29.43\}$.**Interpretation:** The firm breaks even at approximately $Q = 0.57$ (barely above shutdown) and $Q = 29.43$ (the high-output break-even point). For $Q \in (0.57, 29.43)$, the firm makes positive profit; outside this range, it makes a loss. The firm should produce $Q^* = 15$ to maximise profit.